PharmaLeaf · B.Pharm Semester IV · PCI 2026 (NEP 2020)

BP401T Herbal Drug Technology

PharmaLeaf notes for B.Pharm Semester IV (PCI 2026, NEP 2020). Written in easy language, unit by unit in syllabus order, with labelled diagrams, comparison tables, flow charts, examples and memory tricks.

Syllabus at a glance

BP401T is a 3-credit core theory course: 3 lectures per week, 45 hours in total, 75 marks (30 sessional + 45 end-semester).

Unit	Title	Topics	Hours
I	Plant tissue culture and standardized extracts	History, types of culture, nutrition, callus and suspension culture; elicitation, transformation, biotransformation, precursor feeding, somaclonal variation; biomanufacturing (shikonin, paclitaxel), plant factories, oral vaccines; standardized extracts and TLC fingerprints of 10 plants	10
II	Herbal formulations, excipients and cosmetics	Conventional and novel herbal dosage forms (phytosomes, liposomes, nanoformulations); herbal cosmetics; natural excipients	10
III	Quality control and standardisation	WHO, AYUSH and EU guidelines; stability and shelf life; testing methods; DNA markers (rbcL, matK, SCAR); forensic pharmacognosy	7
IV	Herb-drug / food / herb interactions	ADME, CYP450, P-gp; named interaction examples; adverse reactions	10
V	Regulatory requirements	Indian framework for ASU drugs: regulatory bodies, ASU DTAB, ASU DCC, Schedule T, Schedule EI, D&C Act provisions; phytopharmaceuticals and Ayush Aahara	8

Unit I: Plant Tissue Culture and Standardized Extracts (10 hours)

I.I Plant tissue culture as an alternative source of medicine

Plant tissue culture (PTC): *growing* ***plant cells, tissues or organs*** *on a* ***sterile nutrient medium*** *under* ***controlled conditions*** *(aseptic, defined temperature, light and humidity).*

Totipotency: *the ability of a* ***single plant cell to regenerate a whole plant****. It is the basic principle behind PTC (proposed by Haberlandt).*

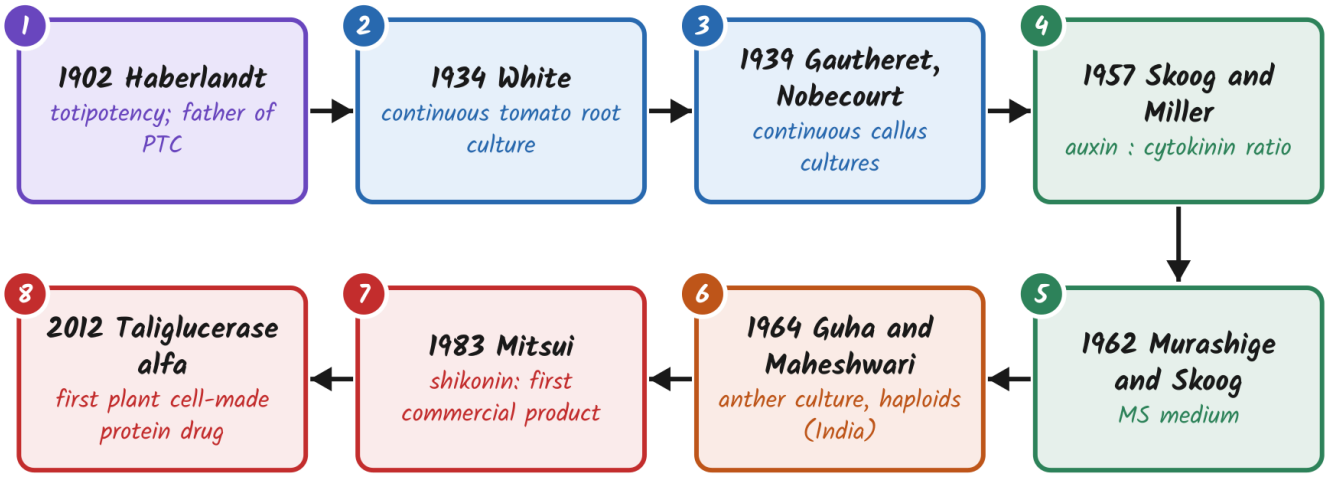
Why PTC as a source of medicines?

Problem with field-grown plants	How tissue culture helps
Slow growth (Taxus trees take decades)	Fast, year-round production in vessels
Variation with season, soil, climate, pests	**Uniform, controlled** quality
Over-harvesting of rare or endangered plants (Taxus, Podophyllum, Rauvolfia serpentina)	**Conservation**: no need to cut wild plants
Pesticides, heavy metals, microbes in crude drugs	**Clean**, contamination-free product
Low content of the useful compound	**Elicitation, precursor feeding, selection of high-yielding lines** increase yield
Political and transport problems with imported plants	Local, factory-based supply
Hard-to-synthesise complex molecules	Cells make them naturally; new compounds by **biotransformation**

Limitations: *high cost, slow growth of plant cells compared with microbes, genetic instability, low yields for many compounds, products often stored inside cells, difficult scale-up (shear sensitivity, aggregation).*

Historical development

Milestones in Plant Tissue Culture

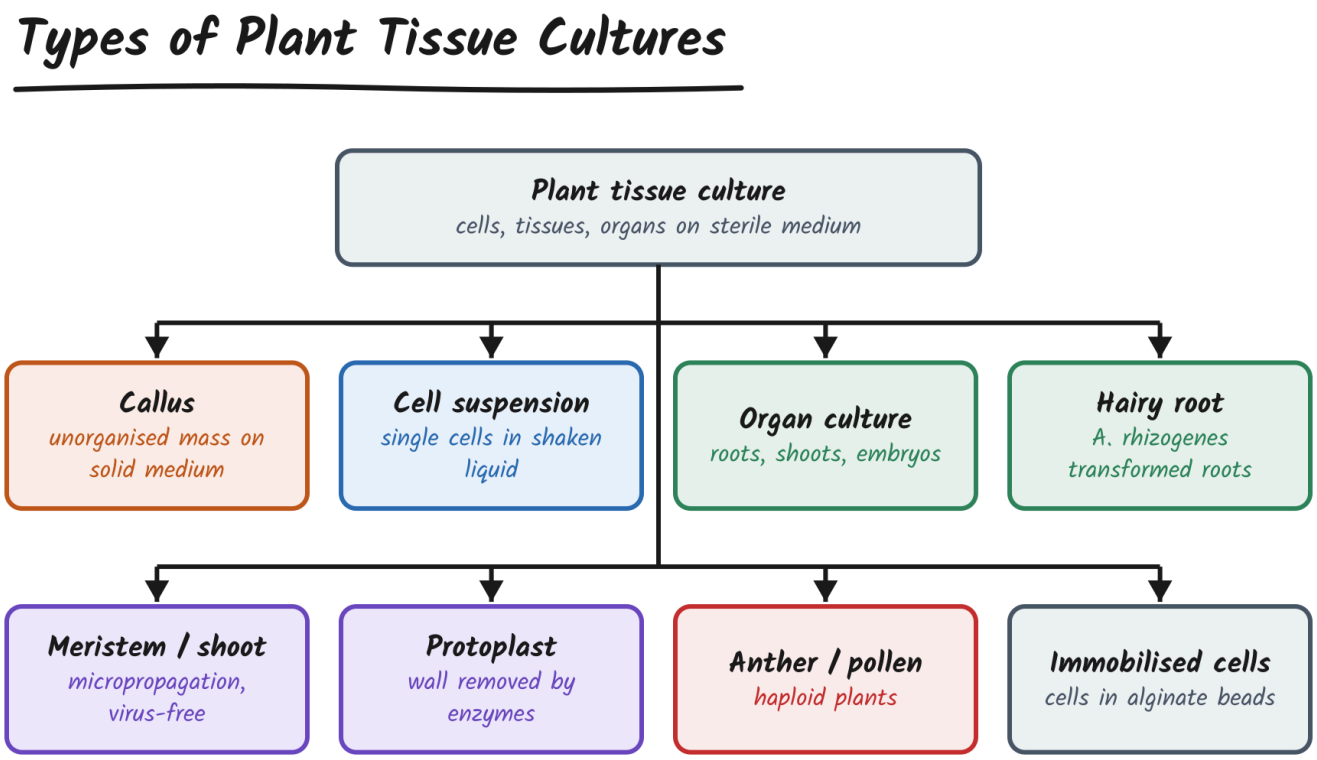


Remember: H-W-G-S-M-G-M: Haberlandt to MS medium to shikonin

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Year	Scientist	Milestone
1902	**Gottlieb Haberlandt**	Proposed **totipotency** and tried to culture single cells; "**father of plant tissue culture**"
1934	**Philip White**	First **continuously growing root culture** (tomato)
1939	**Gautheret, Nobécourt and White**	Independent establishment of continuously growing **callus cultures** (carrot, tobacco)
1955	**Miller and Skoog**	Discovered **kinetin** (a cytokinin)
1957	**Skoog and Miller**	**Auxin : cytokinin ratio** controls root vs shoot formation
1962	**Murashige and Skoog**	**MS medium**: the most widely used culture medium
1964	**Guha and Maheshwari** (India)	**Anther culture**: haploid plants of *Datura innoxia*
1971	Takebe and co-workers	Regeneration of whole plants from **protoplasts**
1983	**Mitsui Petrochemical** (Japan)	**Shikonin**: first commercial secondary metabolite from plant cell culture
1990s–2000s	Phyton and others	**Paclitaxel** by plant cell fermentation
2012	Protalix / Pfizer	**Taliglucerase alfa** (carrot cell culture): first plant cell-made therapeutic protein approved (US FDA)

I.2 Types of cultures



Remember: for metabolites use callus, suspension and hairy roots; for plants use meristem and anther culture

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Type	What is grown	Use
Callus culture	Unorganised mass of dividing cells from an explant on solid medium	Starting point for suspensions, regeneration, metabolite production
Cell suspension culture	Single cells and small clumps in **liquid** medium, agitated	**Large-scale metabolite production** in bioreactors
Organ culture	Roots, shoots, embryos, meristems	Metabolites made only in organised tissue
Hairy root culture	Roots transformed by ***Agrobacterium rhizogenes***	Fast, stable production of root metabolites (tropane alkaloids, ginsenosides)
Shoot / meristem culture	Shoot tips, axillary buds	**Micropropagation**, virus-free plants
Protoplast culture	Cells with wall removed by enzymes (cellulase, pectinase)	Somatic hybridisation, genetic transformation
Anther / pollen culture	Anthers or microspores	**Haploid** plants for breeding